PlanArena: a validated land-use optimizer

Measuring the multidimensional impact of land-use decisions, with stakeholder interests as votes, on official data for ten US cities

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Abstract

Deciding which land uses go where is a location-allocation problem with multidimensional effects: a change alters homes, jobs, access to services, the tax base, emissions and runoff, interacts with neighbouring parcels, and may never be built. US cities decide such changes at public hearings, where staff reports cover a few indicators and residents testify but do not vote. PlanArena is an open land-use optimizer that measures these effects and turns stakeholder interests into votes. It scores any parcel-level change on 11 social, economic and environmental indicators, with sensitivity ranges, in 45 ms (Manhattan) to 0.57 s (New York City); learns from parcel histories how likely a change is; recommends plans within zoning and precedent; and lets resident, environmental, business, developer and fiscal interests vote under nine rules. It covers 4,995,208 parcels in ten US cities from official land-use, plan and zoning data. Tested on observed outcomes, predicted effects of Manhattan changes correlated with recorded homes ($r = 0.840$) and emissions ($r = 0.857$), and redevelopment was ranked out of time with AUC 0.847–0.952. Likelihood weighting brought housing projections from 76–193 to 0.81–2.64 times what was built, and the optimizer found the exact optimum in 93% of small test areas.

Keywords: land-use optimization; location allocation; impact assessment; planning support systems; computational social choice; model validation

1 Introduction

Deciding which land uses go where (housing, jobs, services, open space) is a location-allocation problem [1, 2]. Its effects are hard to measure for four reasons.

- **Many dimensions.** One change alters homes, jobs, access to shops, parks and facilities, the tax base, energy use and runoff, and these trade off against each other [3, 4].
- **Spatial interaction.** Effects spread to neighbouring parcels through walking distance, land-use mix and competition for development [5].
- **Uncertainty.** A permitted change may never be built. In three Manhattan rezonings replayed here, capacity projected 8,423 homes on development sites; 561 were built by 2025.
- **Competing interests.** Residents, developers, businesses, environmental advocates and city finances weigh these effects differently, and no single answer is correct [6].

How decisions are made today. In US cities, rezonings and plan amendments are decided at public hearings.

- **The process.** Planning staff prepare a report, and large rezonings, such as those replayed here, add an environmental review that projects development on identified sites. Residents and advocates testify; the planning commission and the council vote.
- **Testimony is advisory.** Residents are heard but do not vote. Speakers at planning and zoning meetings are disproportionately older, male, long-time homeowners [7], and legally required hearings meet few goals of participation [8–10].
- **Fragmented evidence.** Environmental, fiscal and access effects arrive as separate studies, if at all, and are rarely weighed together.

What computation can measure. Each part of the problem now has a technical tool:

- **parcel-level impact models** calibrated on observed records, which measure many indicators at once;
- **supervised learning on parcel histories**, tested out of time, which estimates how likely a change is [11, 12];
- **graph neural networks** over parcel adjacency, which represent spatial dependence;
- **many-objective evolutionary search** [13], which explores the combinatorial space of plans;
- **computational social choice and bargaining** [14, 15], which turn stakeholder preferences into a decision. Game-theoretic simulation has been applied to collective land-use decisions [16].

Planning support systems (PSS) and urban digital twins aim to combine such tools [17–20]. Three gaps remain:

- **Untested outputs.** Their outputs are seldom tested against outcomes [21–24].
- **Optimization.** It rarely asks whether recommended changes are plausible, or how the choice depends on the decision rule [25–27].

- **Fragmented data.** The data are spread across agencies and schemas, some only as documents: Philadelphia's land-use plan exists only as maps in 17 PDF district plans.

This paper. PlanArena is an open land-use optimizer that measures the impact of any parcel-level change and makes stakeholder interests part of the decision. It contributes:

1. **Measurement.** Any edit is scored on 11 indicators in three pillars, with sensitivity ranges, at interactive speed. Each edit also shows what exists, what zoning permits, what the plan intends and how likely the change is.
2. **Optimization and decision.** The optimizer recommends plans within zoning and precedent. Resident, environmental, business, developer and fiscal interests vote on candidate plans under nine rules, and residents also vote block by block.
3. **Official data for ten cities.** Existing use, plan and zoning for 4,995,208 parcels are on one schema, with a validated method to digitise plans published as vector PDF maps.
4. **Validation of every component** on observed data (Table 3), with design guidance drawn from it.

Scope. Impacts are validated in Manhattan only; elsewhere coefficients are national or regional. Votes rest on assumed utilities, not elicited preferences, and usability with planners is not evaluated.

2 Related work

PSS and urban digital twins. PSS combine suitability, demand projection and allocation for scenario planning [17, 18], often in a geodesign workflow linking representation, change, impact and decision [28]. Their limited use and uncertain usefulness are long-standing concerns [21, 29, 30]. Digital twins extend them to integrated, updatable city models [19, 20, 31–33]. Reviews report technical and organisational barriers [22, 34], and ambitions ahead of demonstrated contributions [23, 24, 35]. PlanArena attaches a test against observed data to each output.

Optimization and collective decisions. Location-allocation models site facilities and uses [1, 2]; multicriteria analysis and multi-objective optimization generate and rank land-use alternatives [3, 4]. Recent work uses deep reinforcement learning [25] and explicit stakeholder interests [26, 27]. Planning problems have no single correct solution [6], so the rule that turns preferences into a choice matters. Bargaining [14] and voting rules [15, 36] have different properties, and bargaining has been combined with land reallocation [37]. Simulated votes are not participation [8]; PlanArena uses them only to show how a choice depends on its rule.

Predicting development, and regulation. Urban change models range from cellular automata [38] and statistical site models [39] to behavioural [40, 41] and agent-based simulation [42], with neighbourhood and temporal effects as priorities [5, 43].

- **Parcel models** capture existing and historical neighbourhood conditions [11], and machine learning complements econometrics [12].
- **Redevelopment** depends on land value relative to the structure [44, 45]. In Chicago, smaller and older buildings with less lot coverage are replaced more often [46, 47].
- **Regulation is not exogenous.** Rezoning follows development opportunity [48], studies differ on upzoning's short-run effect on construction [49–51], and plans are only partly implemented [52–56].
- **Regulation in models.** Recent land-change models integrate zoning and planning rules [57–59] but do not test what they add to prediction on held-out history.

Validation. Land-change predictions should be compared with persistence at several resolutions [60, 61], and their transition probabilities validated directly [62]. Under spatial and temporal dependence, random cross-validation overstates skill [63, 64], and models apply only where new data resemble the training data [65]. Uncertainty from many coefficients is better assessed by varying them jointly [66].

3 The system

A parcel twin holds each city's official layers (Figure 1). Four engines answer the planner's questions:

- the **impact engine**: what would a change do?
- the **likelihood models**: is it likely?
- the **optimizer**: what would do better?
- the **decision module**: who would choose it, and under which rule?

In the Studio, a browser application, a planner paints uses and floor-area ratios on parcels and sees every indicator, flag and recommendation update (Figure 2).

Parcel twin and official layers. The twin covers the nine largest US cities except San Jose, plus Milwaukee (Table 1).

- **Existing land use** is the parcel land use the city or its regional planning agency publishes, refined with assessor records where the layer groups uses.

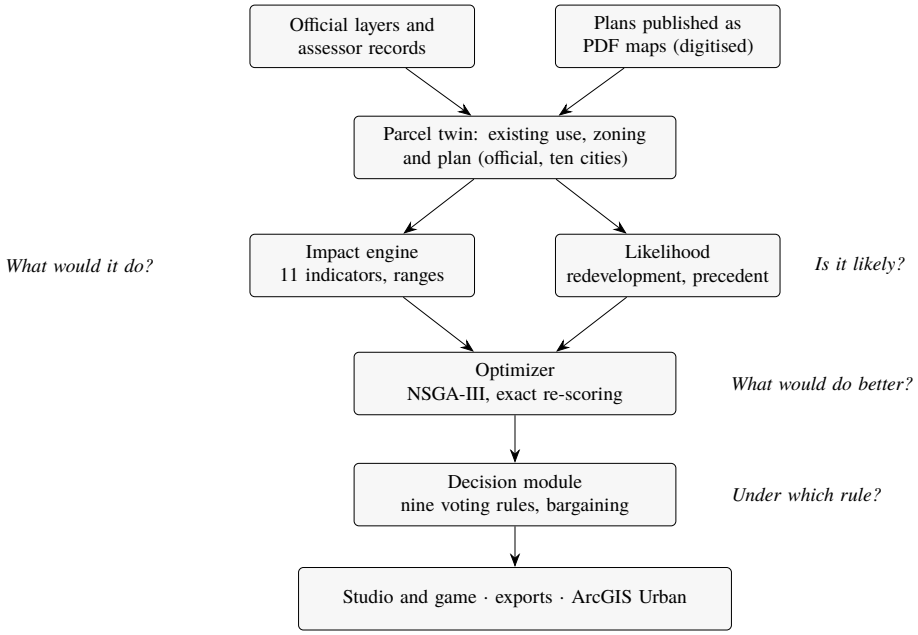


Figure 1: Architecture. Every number the system shows comes from one of these components, each tested on observed data (Table 3).

Table 1: The ten cities: official parcel layers on one schema. Shares are of the city's parcels.

City	Parcels	Existing use (source)	Share	Adopted plan	Share	Zoning	Homes / Census
New York	856,634	MapPLUTO (City Planning)	100%	none citywide	–	100%	1.03
Los Angeles	672,444	SCAG 2020; county assessor	100%	General Plan (SCAG 2019)	100%	100%	0.99
Chicago	606,894	CMAP 2023; county assessor	100%	none citywide	–	96%	1.05
Houston	630,525	City (appraisal codes)	97%	none	–	–	0.95
Phoenix	510,871	County assessor; MAG	100%	General Plan	100%	99%	0.99
Philadelphia	547,410	City Planning Commission; OPA	100%	district plans (digitised PDF)	83%	100%	0.99
San Antonio	438,009	Bexar Appraisal District	100%	adopted area plans	61%	82%	0.99
San Diego	272,948	SANDAG 2025; county assessor	100%	General and community plans	100%	94%	1.02
Dallas	311,431	City 2026; DCAD	100%	forwardDallas! 2.0	99%	79%	0.98
Milwaukee	148,042	MPROP	100%	PDF only (not included)	–	99%	1.06
All	4,995,208						

Existing use is the city's or regional agency's parcel land use, refined by assessor records where the layer groups uses. Homes / Census: homes in the twin over Census 2020 housing units. Houston has no zoning. Milwaukee's area plans are published as PDFs that could not be downloaded by script.

- **The adopted plan** (future land use) and **zoning** are separate layers.
- **Classes.** Explicit crosswalks map every source code to 16 classes: three housing types, mixed use, retail, office, hotel, industry, warehousing, education, health, civic, open space, transportation, parking and vacant land. The classes nest in nine analysis groups.
- **Records.** Floor area, homes, year built and assessed value come from assessor or city records, with estimated fields flagged. Jobs are fitted to LEHD block counts.
- **Provenance.** Inputs are downloaded by script, with URL and SHA-256 hash recorded.
- **Philadelphia's plan**, published only as PDF maps, is digitised (Appendix A): 15 of 17 plans, covering 83% of parcels, with median registration errors of 1.1–7.2 ft, each checked against a shifted-map control.
- **Missing plans.** Where a city publishes no plan as data, the layer is absent, not estimated.

Impact engine. An edit sets a parcel's use and, optionally, its floor-area ratio (FAR). The engine computes the 11 indicators of Table 2 within a ten-minute walk and citywide.

- **Unchanged parcels** keep their observed values, so the baseline reproduces the records exactly.
- **Changed parcels** follow a hybrid delta rule, chosen on observed change (Section 5.2). A density-only change on a consistent record scales the observed composition; a use change removes the observed old use and adds the modelled new one.
- **Derived quantities.** Homes follow from residential floor area and the parcel's floor area per home. Jobs follow from floor-area components and rates fitted to LEHD counts. Emissions follow from energy intensities and emission factors derived from New York's benchmarking data (Local Law 84) and emission limits (Local Law 97). Runoff follows from surface coefficients, and value from the difference between modelled values.
- **Access indicators** are recomputed on a 50 m raster.
- **Ranges.** Every coefficient has a source and a status, and low and high variants give each indicator a sensitivity range.

Indicators combine into pillar scores with weights the planner sets. Scoring a 50-parcel edit, ranges included, takes 45 ms on

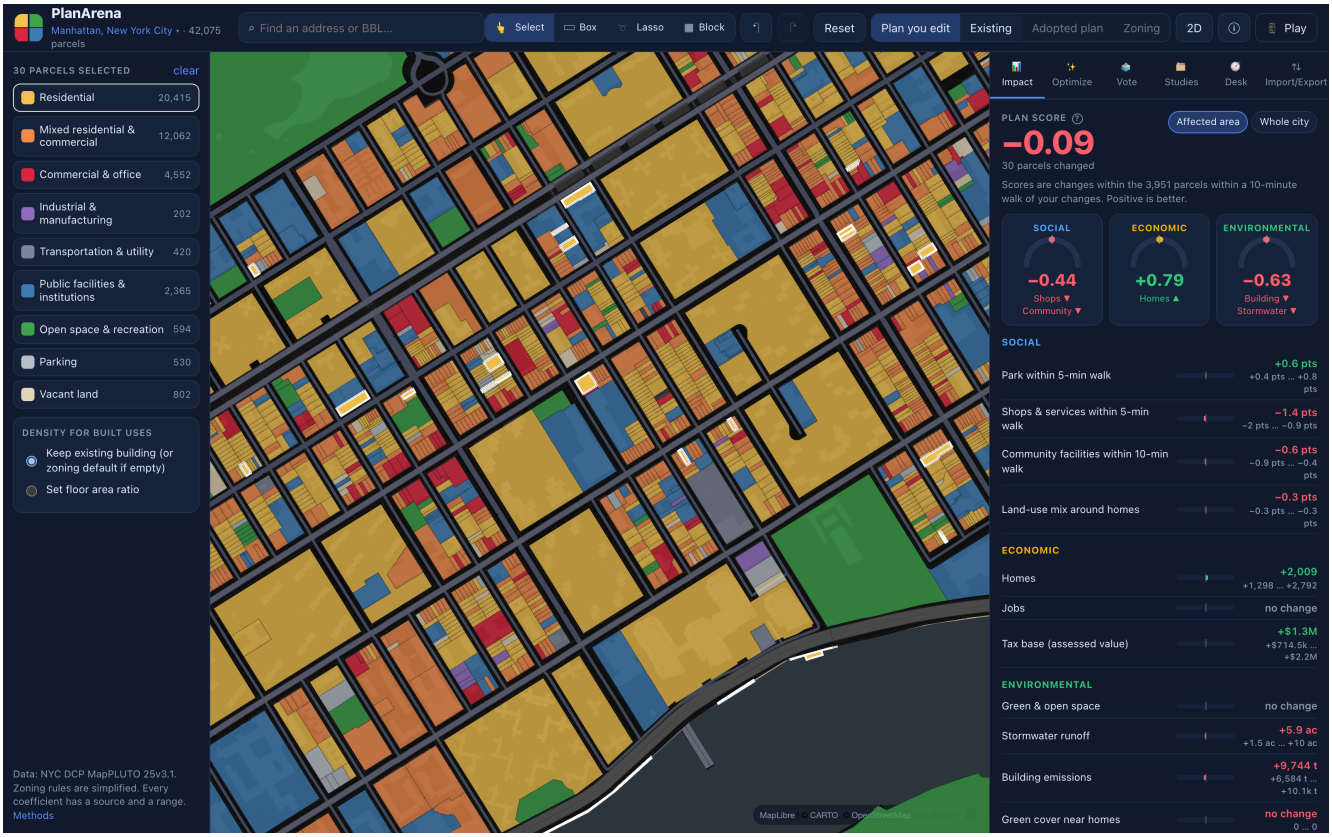


Figure 2: The Studio in Manhattan: thirty parking lots and vacant parcels in East Harlem painted as housing (outlined). Right: plan score, social, economic and environmental pillars, and each indicator's change within a ten-minute walk with its sensitivity range. Left: the use palette. Top: switches between the plan being edited and the city's existing, planned and zoned use.

Manhattan's 42,075 parcels and 0.57 s on New York City's 856,634. Impacts are computed for the nine analysis groups, for which the coefficients are calibrated and validated.

Likelihood, precedent and pattern. Three models judge plausibility.

- **Redevelopment.** A gradient-boosted model (LightGBM), trained only on outcomes known at the decision year, gives each lot's probability of redevelopment within one, three or five years and its likely new use. Its inputs are the lot's use, size, built and permitted floor area, age and land share of value, its neighbourhood and its zoning.
- **Precedent.** A conversion is flagged if buildings of that use and size were not converted to the proposed use in the past decade.
- **Pattern.** A graph neural network over parcel adjacency, combined with LightGBM, describes how well a use fits its surroundings: an existing pattern, not a norm.

Edits that depart from the plan or from what zoning allows are flagged (Figure 3).

Optimizer. The optimizer maximizes the weighted pillar score, minus a penalty per conversion, under a budget of changes. Zoning is a hard constraint, conversions without precedent are skipped, and moves can be weighted by likelihood. Two searches are offered:

- iterated best response over first-order move estimates, with exact re-scoring of the best moves;
- NSGA-III [13], which returns a Pareto set across the pillars.

Both report the best plan found, not a proven optimum.

Decision module. Choices are contested [6], so the module shows how the chosen plan depends on the decision rule.

- **Agents.** Five stakeholder agents (residents, business, developers, environmental and city-finance interests) score candidates from assumed utilities over the indicators.
- **Rules.** Nine voting rules are compared with the Nash bargaining solution [14] and a residents' referendum by block. Schulze's method [36] decides.
- **Reports.** Whether a Condorcet winner exists, whether one insincere ballot could change the outcome, and the winner's robustness to coefficients and seats.

Unlike hearing testimony, these interests are votes: each stakeholder holds seats on a simulated council, and residents also vote

Table 2: Indicators computed for every edit, and how each is validated (Manhattan).

Indicator	Pillar	Unit	Better	Validation
Homes	economic	units	higher	observed change
Jobs	economic	jobs	higher	existing (tract, block)
Tax base (assessed value)	economic	USD	higher	observed change
Park within 5-min walk	social	% of residents	higher	not validated
Shops & services within 5-min walk	social	adequacy 0-1	higher	not validated
Community facilities within 10-min walk	social	adequacy 0-1	higher	not validated
Land-use mix around homes	social	index 0-1	higher	not validated
Green & open space	environmental	acres	higher	direct accounting
Stormwater runoff	environmental	impervious-equiv. acres	lower	not validated
Building emissions	environmental	tCO2e / yr	lower	existing and observed change
Green cover near homes	environmental	% of land within 400 m	higher	not validated

Unchanged parcels keep their observed values; changed parcels follow the delta equations. Each indicator carries a sensitivity range from low and high coefficient variants. Access, mix, green cover and runoff are recomputed from the resulting state but have no observed after-change data to test against.

Table 3: Validation design: each component of the system and its test against observed data.

Component	Question	Data	Section
Existing conditions	Does the twin reproduce what is there?	LL84 energy, LODES jobs, recorded homes	5.1
Planning calculations	Does it reproduce a review's projection from the review's own inputs?	FEIS site tables (two rezonings)	5.1
Effects of change	Given an observed change, does it predict the effect?	MapPLUTO 2015 vs 2025	5.2
Likelihood	Does it rank lots that redevelop, forward in time?	Rolling origins 2016–2025	5.3
Official layers	What do zoning and plans add, in other cities?	Seven city panels	5.4
Scenario projections	Are projected homes close to what was built?	Four cities; three rezonings	5.5
Optimizer	How close is the best plan found to the optimum?	Exhaustive search; 12 districts	5.6
Realism	Are recommended changes plausible?	2020 plans vs 2025 outcomes	5.6

block by block. Utilities are editable assumptions, and the output is a simulated scenario, not public support [8] (Figure 4).

Interface. Planners can edit parcels by click, box, lasso or search, or upload plans as tables or GIS files, which are reviewed before they are applied. Studies are stored with a hash-chained audit log. Exports include a staff-report draft, CSV and GeoJSON files, and the provenance of every coefficient. A phone version presents the engine as a game. Any city or district exports to ArcGIS Urban: space-use, zoning and land-use types, boundaries, and parcels with a PlanArena scenario.

4 Validation design

Each component is tested on observed data (Table 3). Predictions are trained on one period and tested on a later one; uncertainty comes from resampling spatial tiles [67], never from random splits [63]. The cross-city test uses seven cities with parcel histories (Appendix Table 11).

- **Development** is defined as each source records it: a new building or large floor-area increase (Los Angeles, Dallas, New York), a new building (Milwaukee), a change of use class between land-use inventories (Phoenix, San Diego), or a new-construction permit (Chicago).
- **Blocks.** Four feature blocks are added in turn: existing conditions (E), context within 400 m (C), zoning (Z) and plan (P).
- **Leakage control.** Where a current zoning or plan layer has dated changes, a conservative variant drops lots whose zoning or plan dates from the window's start or later.

5 Results

5.1 Existing conditions and planning calculations

The twin reproduces Manhattan's existing conditions.

- **Emissions.** Modelled emissions match 6,234 benchmarked buildings in a year not used for fitting (log–log $r = 0.807$, median error 31%, total 1.04 times metered).
- **Jobs** correlate with LEHD counts at tract level ($r = 0.843$) and block level ($r = 0.727$).
- **Homes.** Given observed use and floor area, homes on 295 lots developed in 2015–2025 total 1.04 times the recorded count ($r = 0.903$).
- **Environmental reviews.** Given two rezoning reviews' own sites, floor area and uses, PlanArena reproduces their projected homes at 0.86 and 1.00 times the review figures.

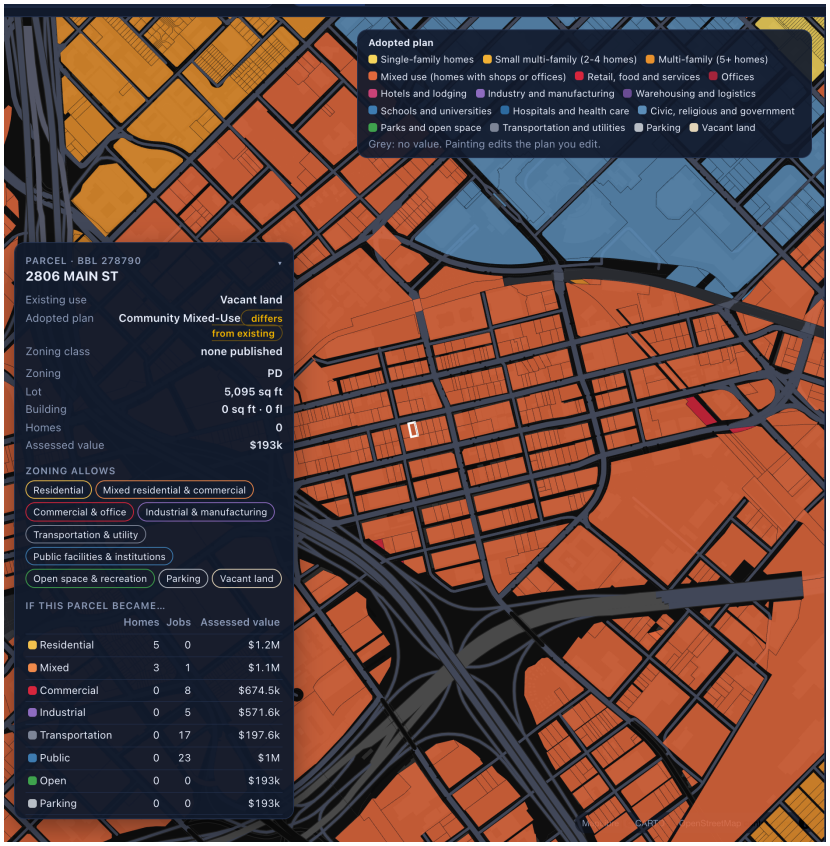


Figure 3: Parcel inspector on the adopted-plan layer (Dallas). For a vacant lot the plan designates Community Mixed-Use, it lists existing use, plan designation (flagged as differing), zoning, the uses zoning allows, and what each new use would yield in homes, jobs and assessed value.

Table 4: Effects of observed change, Manhattan 2015–2025: given the 2015 state and the observed new use and floor area, the predicted change against the recorded change (Pearson r; homes: predicted over observed total).

Method	Changes	Homes r	Homes total	Value r	Emissions r
PlanArena	all	0.840	1.31×	0.870	0.857
	use change	0.872	1.29×	0.901	0.857
Mean change of the same transition	all	0.427	0.99×	0.148	0.090
	use change	0.435	0.97×	0.233	0.090

The baseline is cross-fitted: each change takes the mean observed change of other changes of the same type and size class. Emissions are tested on the changed buildings in the energy benchmarking data. Neither method predicts job change at block level.

5.2 Effects of observed change

For 3,350 Manhattan lots that changed in 2015–2025, each was given its 2015 state and observed new use and floor area (Table 4). Against the mean change of the same transition type:

- **homes**: predicted change correlates with recorded change at $r = 0.840$, against 0.427;
- **assessed value**: $r = 0.870$, against 0.148;
- **emissions** of 194 benchmarked buildings: $r = 0.857$, against 0.090.

The engine overestimates total homes added (1.31 times observed), which the baseline matches by construction. Neither predicts job change by block (engine $r = 0.068$). Access, mix, green cover and runoff have no after-change data, and the Studio marks them as not validated.

5.3 Likelihood of redevelopment

Trained on 2015–2020 outcomes and tested on 2020–2025, the redevelopment model reaches:

- AUC 0.847, against 0.754 for the best simple baseline and 0.809 for a graph neural network;
- 26% redeveloped among its top 1% of lots, against a base rate of 1.35%;
- expected calibration error 0.011.

Across 15 rolling origins (2016–2025; horizons of one, three and five years), its out-of-time AUC is 0.847–0.952, above the best baseline at every origin.

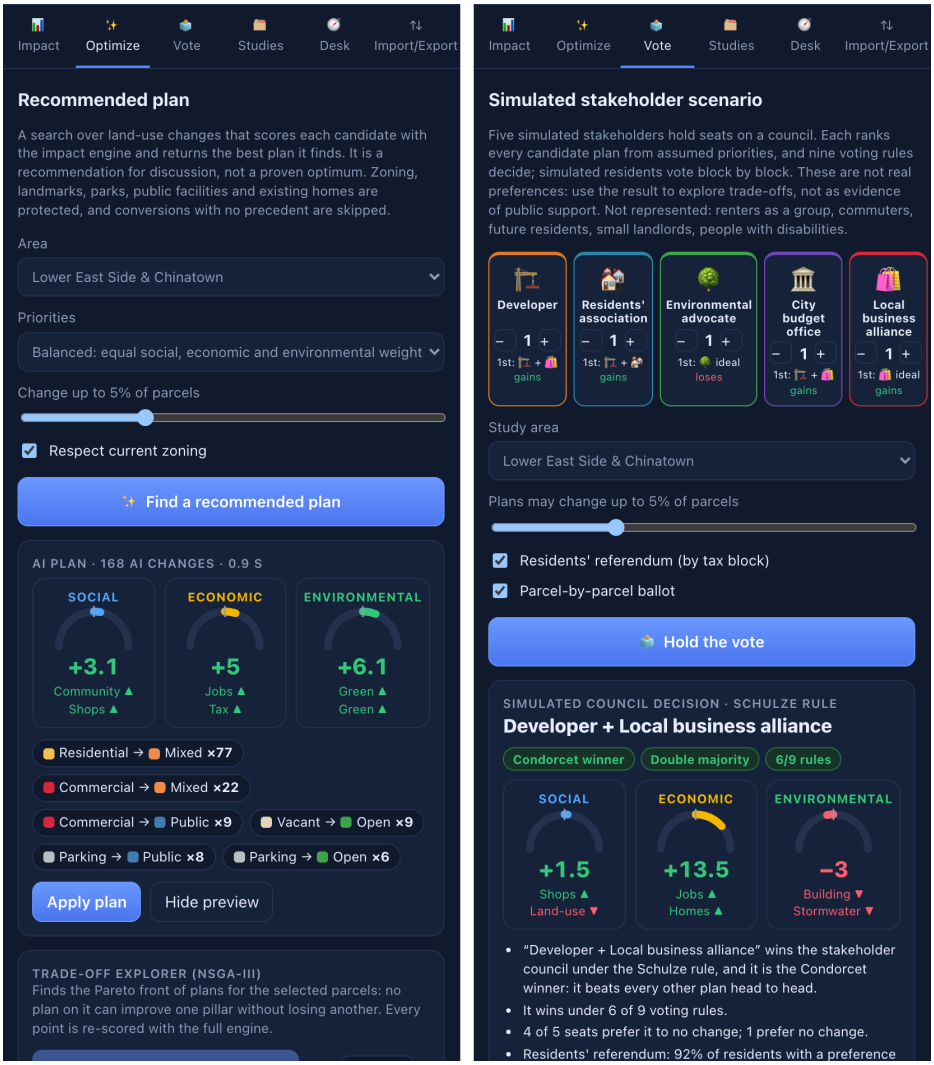


Figure 4: Optimizer and decision module for one Manhattan district. Left: the recommended plan, its change in each pillar and its conversions. Right: the simulated council's decision under Schulze's rule, whether the winner is a Condorcet winner with a double majority, and how many of the nine rules it wins.

5.4 What zoning and plans add, across seven cities

Existing conditions alone rank developed lots with out-of-time AUC 0.824–0.911 in six cities, and 0.571 in San Diego, whose one-year window recorded few changes (Appendix Table 12). Adding blocks:

- **Context** adds the most: +0.020 to +0.116, significant in five cities (Table 5, Figure 5). Phoenix is the exception: its land-use series starts in 2022, so recent change is unknown in its training window.
- **Zoning** carries its own signal. Without context, it raises AUC significantly in five of seven cities, and developed lots were 2.1–9.5 times as likely as others to have a zoning class outside their existing use group.
- **Overlap.** Most of that signal is shared with existing conditions and context: developed lots were also 3.5–19 times as likely to be vacant. Over context, zoning adds a significant gain in four of seven cities and the plan in three of four, at most +0.021 outside Phoenix.
- **Phoenix.** Zoning adds +0.049 and the plan +0.054, and 45% of developed lots were vacant.
- **Conservative variant.** The plan gain in San Diego and the zoning gain in Chicago do not survive it: those gains came from lots rezoned or re-planned during the study.

The layers matter more for what developing lots become (Table 6).

- **Zoning class.** In all six cities with a destination, it names the new class correctly more often than the existing use does, by 12–64 percentage points. In Phoenix and San Diego, development is a change of class by definition, so the existing use is never right at class level.
- **Plan class.** In Dallas it is right for 80% of developed lots, against 61% for zoning.
- **Transfer.** Models fitted on other cities predict worse than the city's own in four of five cities (Appendix Table 13).

Table 5: Gain in out-of-time AUC from adding a block, with 95% bootstrap intervals over 2 km tiles. Bold: the interval excludes zero.

City	C over E	Z over E+C	P over E+C	Z, conservative	P, conservative
Los Angeles	**+0.028 [0.020, 0.036]**	+0.002 [−0.001, 0.005]	0.000 [−0.004, 0.003]		
Dallas	**+0.020 [0.015, 0.025]**	**+0.004 [0.002, 0.005]**	**+0.002 [0.001, 0.003]**		
New York	**+0.023 [0.018, 0.029]**	**+0.004 [0.003, 0.007]**	–		
Milwaukee	−0.015 [−0.079, 0.048]	+0.029 [−0.031, 0.075]	–		
Phoenix	−0.066 [−0.139, 0.001]	**+0.049 [0.009, 0.091]**	**+0.054 [0.005, 0.100]**	**+0.055 [0.017, 0.088]**	**+0.041 [0.004, 0.073]**
San Diego	**+0.116 [0.053, 0.170]**	−0.003 [−0.022, 0.018]	**+0.021 [0.003, 0.039]**	−0.009 [−0.029, 0.015]	−0.011 [−0.033, 0.007]
Chicago	**+0.063 [0.049, 0.078]**	**+0.004 [0.001, 0.007]**	–	−0.001 [−0.005, 0.003]	–

Conservative: in each window, lots whose zoning or plan dates from the window's start or later are left out; possible where a current layer records those dates (Phoenix, San Diego, Chicago). Dallas's current layers carry no dates, so its zoning and plan gains are upper bounds.

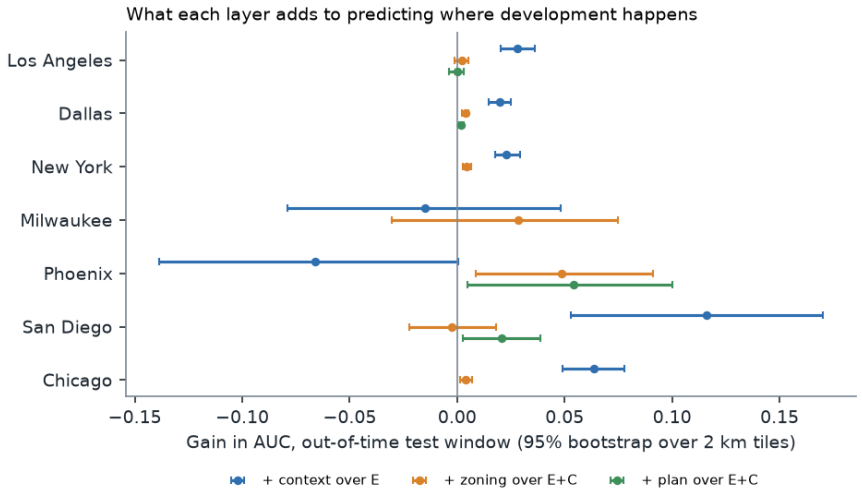


Figure 5: Gain in out-of-time AUC from each block, with 95% tile-bootstrap intervals.

5.5 Scenario projections

Capacity overstates near-term housing by two orders of magnitude (Table 7).

- **Build-out.** In four cities, zoning and plan build-out exceed the homes added in the test window 76- to 193-fold. Part of the gap is definitional: build-out is an end state, not a two-year flow.
- **Likelihood weighting** brings totals to 0.81–2.64 times the observed.
- **Trend.** A five-year trend ranks districts better in all four cities: Spearman 0.51–0.81, against −0.37–0.55.
- **Manhattan rezonings.** Capacity projected 8,423 homes on development sites, or 5,207 pro-rated to 2025; the likelihood-weighted projection gave 585, against 561 built (Table 8, Figure 6). The observed count lies in the model's 95% range in two of three rezonings.

5.6 Optimizer and plausibility

Optimality. On 20 small areas where every plan with up to two changes can be enumerated, first-order search found the exact optimum in 73%, and with exact re-scoring in 93% (median 0.28 s). At district scale (Table 9), exact greedy search found the best plan in all 12 districts but took a median 25.87 s. First-order search with re-scoring reached a median 97.4% of that score in 1.14 s, the speed used in the Studio.

Plausibility (Table 10).

- **Without realism layers,** 17% of recommended changes had no precedent.
- **With the likelihood layer,** 21% of recommended parcels were redeveloped by 2025, against 12% without it, 1.4% of all lots and 5.2% of the likeliest fifth.
- **The cost** is a lower plan score: 2.73 against 8.27. The Studio shows both plausibility and score.

5.7 Decision rules

In the simulated scenario for 12 Manhattan districts:

- the nine rules never all agree, and a Condorcet winner exists in seven;
- the residents' referendum picks the council's plan in one district, and the council's winner is never the Nash bargaining solution;
- one insincere ballot could change the Schulze result in six districts;
- the winner survives a median 76% of joint perturbations of coefficients and seats, and 54% of redrawn stakeholder weights.

Table 6: What developed lots became: share named correctly, as one of 16 classes / one of nine groups.

City	Developed	No change	Zoning class	Plan class	Model (blocks)
Los Angeles	1,883	57% / 79%	70% / 87%	70% / 87%	74% / 89% (E+C+Z+P)
Dallas	3,910	37% / 40%	61% / 65%	80% / 89%	86% / 95% (E+C+Z+P)
New York	4,002	24% / 37%	49% / 65%	–	54% / 70% (E+Z)
Milwaukee	78	18% / 18%	37% / 77%	–	63% / 76% (E+Z)
Phoenix	1,835	0% / 35%	64% / 74%	47% / 62%	75% / 81% (E+C+Z+P)
San Diego	202	0% / 52%	14% / 74%	27% / 70%	64% / 75% (E+C+Z+P)

In Phoenix and San Diego development is a change of class (to another group, or to more homes), so "no change" is never right at class level there. Models are cross-fitted over 2 km tiles among the developed lots. Chicago's permits do not record the new use.

Table 7: Homes added in the test window: projection over observed total (Spearman correlation across districts).

City	Observed	Zoning build-out	Plan build-out	Likely × zoning	Likely × plan	Five-year trend
Los Angeles	22,853	193× (0.10)	189× (0.24)	0.81× (0.46)	0.81× (0.52)	0.59× (0.70)
Dallas	10,926	141× (0.00)	127× (0.34)	2.64× (0.00)	1.69× (0.30)	1.25× (0.72)
New York	116,776	76× (0.34)	–	0.96× (0.55)	–	0.47× (0.81)
Milwaukee	3,798	115× (−0.23)	–	0.97× (−0.37)	–	0.32× (0.51)

Build-out: every lot at the homes per acre that the 90th percentile of lots of its zoning (or plan) class reaches in the city today. Likely: the build-out increment times the out-of-time probability of development (blocks E+C+Z+P where available). Trend: homes completed in the five years before the window, scaled to its length.

These describe the simulated scenario, not public opinion. They make explicit what a hearing leaves implicit: resident and environmental interests are inputs to the decision, and a recommendation should be reported with its rule and with how it changes under others.

6 Discussion

What a planner gets. For any proposed change, PlanArena shows:

- its effects on 11 indicators, with ranges;
- whether zoning and the plan allow it, and how likely it is;
- what the optimizer would recommend instead;
- how the choice depends on the decision rule.

Each output has a stated test on observed data (Table 3), and untested indicators are labelled. For one system, this closes the gap between claimed and demonstrated contributions of PSS and urban twins [21, 23, 24].

Design guidance.

1. **Use each layer for the question it answers.** Zoning and plans add little to predicting *where* development happens once existing conditions and context are known. They help predict *what* it becomes and bound *how much* a site can hold. This is consistent with rezoning following opportunity [48], upzoning's varying short-run effects [49–51] and partial plan implementation [53, 55].
2. **Report likelihood-weighted projections beside capacity,** and a trend beside both. Capacity overstates near-term housing by orders of magnitude, and the trend ranked districts better than any projection.
3. **Constrain optimization by plausibility and show the cost.** Realism layers change recommendations and lower the objective.
4. **Report decisions with their rule.** No rule agreed with all the others in any district of the simulated scenario.

Limitations.

- **Impacts** are validated only in Manhattan; in the other nine cities they are unvalidated estimates from national or regional coefficients.
- **Untested indicators.** Job change by block, and access, mix, green cover and runoff after a change, have no observed test.
- **The cross-city test** uses short windows (one or two years) and city-specific outcome definitions. In Dallas, plan and zoning are undated current layers, so its gains are upper bounds.
- **Assumptions.** The optimizer is heuristic, and the votes rest on assumed utilities.
- **Not studied.** Usability with planners is not evaluated, and causal effects of zoning or plans are not estimated.

7 Conclusion

Land-use and location decisions have multidimensional, spatially interacting and uncertain effects, yet they are decided at hearings where those effects are summarised in a few indicators and residents do not vote. PlanArena, a land-use optimizer, measures the effects of any parcel-level change on social, economic and environmental indicators, with what is permitted, planned, likely and recommended, and turns stakeholder interests into votes. It runs on official data for ten large US cities at

Table 8: Three Manhattan rezonings: homes on development sites by 2025, observed and projected.

Rezoning (adopted)	Sites	Built by 2025	Capacity	Pro-rated to 2025	Likelihood-weighted (95%)
East Harlem (2017)	127	288	4,117	3,294	234 (79–489)
Inwood (2018)	47	270	2,859	1,334	213 (0–652)
SoHo/NoHo (2021)	77	3	1,447	579	138 (8–304)
All three		561	8,423	5,207	585

Capacity is full build-out by the review's build year; pro-rated scales it linearly to 2025. The likelihood comes from a redevelopment model trained only on outcomes known by the adoption year.

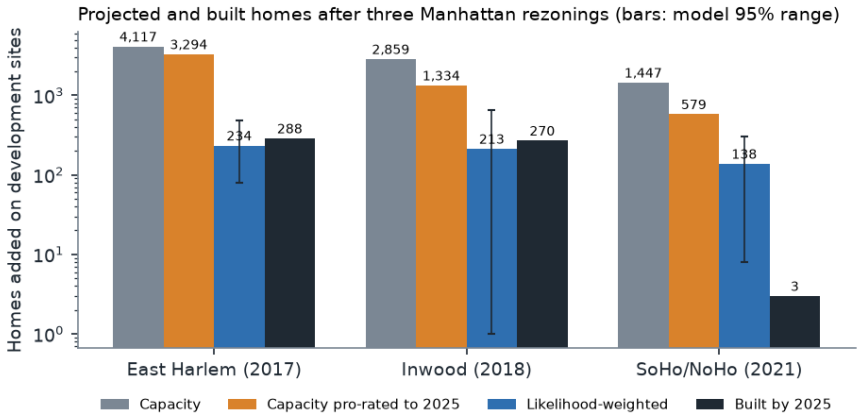


Figure 6: Homes on development sites after three Manhattan rezonings: capacity, pro-rated capacity, likelihood-weighted (with 95% range) and built by 2025 (log scale).

interactive speed, and every component is tested against what happened. The tests also shape the design:

- regulatory layers inform what development becomes and how much it could hold, not where it happens;
- likelihood-weighted projections are far closer to what was built than capacity, while trends rank districts better;
- recommendations change when they must be plausible, and with the rule used to choose.

Code, data manifest and results are open, so the system and its tests extend to other cities.

Data and code availability. The code, the result files behind every number, the figures and a manifest of every input (URL, retrieval date, SHA-256) are in the PlanArena repository. `make reproduce` rebuilds data and results from public sources, and `planarena/tools/screenshots.py` regenerates the Studio figures. Each city's ingestion report documents coverage, crosswalks and estimated fields. Owner and mailing fields in assessor records were never requested or read. No human participants were involved.

Declaration of AI assistance. [To be confirmed by the author.] The author used Claude (Anthropic) to write and test code, run analyses and draft text. The author designed the study, reviewed code and results, read the cited sources and takes responsibility for the content. Every reference was checked against the record its DOI or ISBN resolves to.

A Digitising plans published as PDF maps

Philadelphia2035's land-use recommendations exist only as "Proposed Land Use" maps in 17 district plans covering 18 districts. Each lot is a filled vector shape. The maps are converted in six steps:

1. **Legend.** Swatch colours and labels are read.
2. **Colour matching.** Map colours are matched one to one to legend colours by CIELAB ΔE with a Hungarian assignment, refusing pairs with $\Delta E > 45$.
3. **Visibility.** Only shapes the rendered page shows are kept, since maps are often clipped to the district, and only the largest map frame on the spread.
4. **Registration.** A similarity transform is fitted. The rendered footprint is matched as an image to the district's parcels and outline (FFT translation, Dice score), then refined by iterative closest point [68–70].
5. **Classification.** Each parcel takes the category of the smallest shape containing its interior point.
6. **Checks.** A map is used only if the median parcel-to-shape distance is at most 10 ft, and its plan agrees with existing use clearly more often than the same map shifted 300 ft.

Results:

- **Coverage.** Maps from 15 of 17 plans passed, covering 16 districts and 83% of parcels.

Table 9: Optimizer at district scale (12 Manhattan districts, up to ten changes): share of the best plan any method found.

Search	Median share of best	Districts best	Median seconds
surrogate top moves	92.6%	0	0.02
first order	96.4%	1	0.13
first order plus exact polish	97.4%	1	1.14
greedy exact	100.0%	12	25.87
simulated annealing	94.8%	0	7.54
random restarts 50	65.9%	0	0.94

Table 10: Realism layers: plans recommended on the 2020 snapshot of every Manhattan district, and what happened by 2025.

Optimizer setting	Parcels	No precedent	Redeveloped by 2025	Plan score
no realism layer	529	16.6%	11.9%	7.52
precedent filter (default)	544	0.0%	12.1%	8.27
precedent filter + likelihood layer	118	0.0%	21.2%	2.73

For reference, 1.4% of all lots and 5.2% of the likeliest fifth were redeveloped. The plan score is the optimizer's own objective.

- **Registration.** Median errors were 1.1–7.2 ft (median 3.2 ft).
- **Shifted-map control.** The digitised plans agree with existing use on 60%–90% of parcels, against 42%–66% for shifted maps.
- **Failures.** The maps of South and West Park also draw neighbouring districts and failed registration.

Earlier work addresses scanned raster maps [71–73]; vector exports allow shapes to be read directly.

B Additional results

Table 11: Test panels. Models are fitted on the training window and scored on the test window.

City	Development	Source	Train	Test	Lots	Developed	Rate
Los Angeles	new building or floor area +25%	county assessor rolls	2021–2023	2023–2025	667,844	2,222	0.33%
Dallas	new building or floor area +25%	appraisal district files	2022–2024	2024–2026	291,841	3,916	1.34%
New York	new building or floor area +25%	MapPLUTO	2021–2023	2023–2025	856,901	4,064	0.47%
Milwaukee	new building	MPROP	2021–2023	2023–2025	147,539	85	0.06%
Phoenix	change of use group	MAG land-use inventory	2022–2024	2024–2026	503,635	1,835	0.36%
San Diego	change of use group	SANDAG land-use inventory	2023–2024	2024–2025	272,281	202	0.07%
Chicago	new-construction permit	building permits	2021–2023	2023–2025	606,294	1,548	0.26%

Lots already under construction at the start of a window are left out. Floor area +25% also requires +2,000 sq ft. Change of use group: the lot changes to a built use of another of the nine groups, or gains homes.

Table 12: Where development happens: out-of-time AUC by block set. E existing conditions, C neighbourhood context, Z zoning, P adopted plan; – where the city has no such layer.

City	E	E+C	E+C+Z	E+C+P	E+C+Z+P	Best set (any)
Los Angeles	0.864	0.892	0.894	0.892	0.894	E+C+Z+P
Dallas	0.898	0.918	0.922	0.920	0.923	E+C+Z+P
New York	0.892	0.914	0.919	–	–	E+C+Z
Milwaukee	0.911	0.896	0.924	–	–	E+Z
Phoenix	0.859	0.793	0.842	0.847	0.853	E+Z+P
San Diego	0.571	0.687	0.684	0.707	0.703	E+C+P
Chicago	0.824	0.888	0.891	–	–	E+C+Z

Best set: the highest AUC among all fitted combinations, including those without context.

Table 13: Transfer: AUC of a model fitted on the other cities' training windows / the city's own model, on the city's test window.

City	E	E+C	E+C+Z	E+C+Z+P
Los Angeles	0.719 / 0.864	0.817 / 0.892	0.828 / 0.894	0.837 / 0.894
Dallas	0.844 / 0.898	0.877 / 0.918	0.878 / 0.922	0.877 / 0.923
New York	0.825 / 0.892	0.874 / 0.914	0.868 / 0.919	0.866 / 0.919
Milwaukee	0.959 / 0.911	0.965 / 0.896	0.966 / 0.924	0.967 / 0.924
Chicago	0.742 / 0.824	0.808 / 0.888	0.804 / 0.891	0.822 / 0.891

Cities whose outcome is a building or a permit. A missing plan layer is coded as unknown.

Table 14: Out-of-time average precision and precision among the top 1% of lots.

City	Base rate	AP, E	AP, all	All blocks	Top 1%, E	Top 1%, all
Los Angeles	0.33%	0.139	0.147	E+C+Z+P	11.2%	12.8%
Dallas	1.34%	0.115	0.237	E+C+Z+P	12.6%	29.6%
New York	0.47%	0.073	0.122	E+C+Z	11.1%	15.1%
Milwaukee	0.06%	0.024	0.031	E+C+Z	2.3%	2.8%
Phoenix	0.36%	0.091	0.054	E+C+Z+P	9.7%	8.4%
San Diego	0.07%	0.002	0.002	E+C+Z+P	0.7%	0.6%
Chicago	0.26%	0.013	0.043	E+C+Z	2.4%	6.1%

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